

Harm Reduction Program Efficacy: Advocacy for Improved Data

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Harm reduction programs such as syringe services programs have consistently been shown to improve health outcomes for users of injected drugs. Addiction is a disease which is growing in the United States and which can affect not just its sufferers but also addicts' loved ones and support networks, as well as seriously impacting hospitals both financially and in terms of capacity. The goal of harm reduction programs is to minimize the negative health outcomes of drug use by minimizing infectious disease transmission and users' dependencies on unsafe resources.

Business Problem

Harm reduction is a contentious topic in the fields of social welfare and medicine. Though many studies have shown that it improves health outcomes (Pridgen et. al), a long-term goal would be to create a concentrated, localized study and outcome model to specifically support the implementation and funding of local programs, such as the North Carolina Harm Reduction Coalition (NCHRC). The goal of this research project is to do initial exploratory research into the data available from the Centers for Disease Control and Prevention (CDC) and determine its usefulness for building a model that can predict public health outcomes based on harm reduction service availability.

If a model can be developed that shows clear improvement in state health outcomes, this would be a massive boon to grant- and support-seeking. The data would be synthesized into persuasive materials that could be implemented into grants and donor presentations.

Data Explanation

Data is here primarily sourced from CDC's WONDER (Wide-ranging ONLINE Data for Epidemiologic Research) program to find sources of death by state, narrowed down to drug-

caused deaths (which fall into many individual ICD-10 codes; see below). Auxiliary data will be taken from the amFAR Opioid & Health Indicators Database, including information about syringe exchange program legality and availability. The amFAR data is aggregated from other sources as available and serves as a repository of archived, related aggregate data.

From the fully-prepared dataset, the relevant data is:

- “drugdeathrate”: “The rate of death from drug poisoning, including both illicit and prescription drugs, per 100,000 population. Drug poisoning data are from CDC's Detailed Cause of Death data and includes ICD-10 codes X40-X44, X60-X64, X85, and Y10-Y14” (amFAR). I leveraged the rates from years 2014, 2017, and 2019, which are the available data that match other collection years.
- “drugdep”: “Percent of total population aged 12 or older reporting illicit drug dependence or abuse in the past year from the National Survey on Drug use and Health (NSDUH).” Retrieved 2014 (ibid).
- “SSP_legality”: “[W]hether states have specifically authorized syringe service programs statewide in legislation. States where local units have interpreted state laws to allow syringe access services or where no law prohibits syringe exchange are considered to permit syringe exchange programs. States with pilot programs, those where syringe exchange would require legislative action and/or supportive interpretation of local laws, or those with pending syringe exchange legislation are considered illegal.” This data was collected from the HIV Prevention Justice Alliance in 2021 (ibid).
- “num_SSps”: “Total number of syringe exchange programs [available in a given state]. The directory of syringe exchange programs is aggregated by self-reporting to the North American Syringe Exchange Network (NASEN) and is therefore unlikely to be a

comprehensive and complete list of all syringe exchange programs.” Collected from NASEN in 2023 and 2024 (ibid).

- “med_meth”: “[W]hether states reported Medicaid coverage of methadone. Methadone i[...] has been shown to decrease the probability of overdose death, as well as lower rates of infectious disease.” Sourced from Kaiser Family Foundation (KFF) in 2017 (ibid).
- “FQHC_fac”: The number of FQHCs in a state. “Federally Qualified Health Centers (FQHCs) are community-based health service providers funded under HRSA's Health Center Program. The program sets requirements for providing comprehensive health care services including primary, preventative, mental health, and substance abuse services regardless of an individual's ability to pay or their health insurance status.” Sourced from Human Resources and Services Administration (HRSA) in 2021 (ibid).

Methods

Based on the available data, I conducted an exploratory analysis and comparison of outcome metrics against predictive metrics with the goal of identifying predictive metrics which could produce a viable predictive model of public health outcomes. However, preliminary exploratory analysis indicated that no predictive metrics would be suitable for the model. Therefore, the scope of this project has been limited to exploratory research into the available data.

During data preparation, I determined that time-series data was not feasible for this project, given that for most predictive indicators time-series data was not available. Where possible, I have matched predictive and outcome values as close as possible in year (e.g. death rates in 2014 compared against drug dependency rates in 2014).

Analysis

Drug-related death rates and drug dependency rates were used as target variables for this

exploration, and other variables examined as potential predictors. The two target variables are, themselves, positively correlated in 2014, the only year for which both datasets are available.

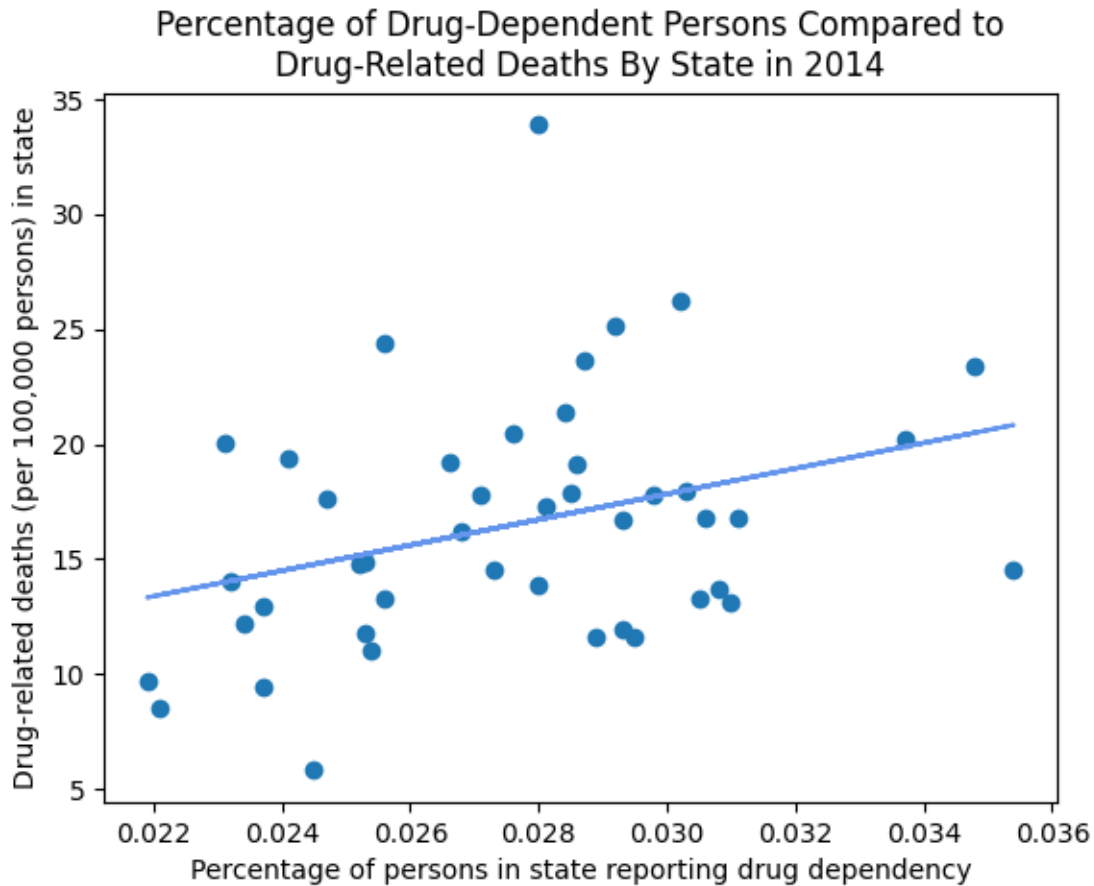


Figure 1: Percentage of drug-dependent persons in a state compared to drug-related deaths (per 100,000 persons) in a state in 2014, showing a positive correlation with a slope of ~556.

I then compared the drug death rate target variable and the drug dependency target variable against the available categorical variables: syringe exchange program legality by state and statewide Medicaid coverage of methadone treatment. Both of these indicated a positive relationship between target variable and legality/coverage.

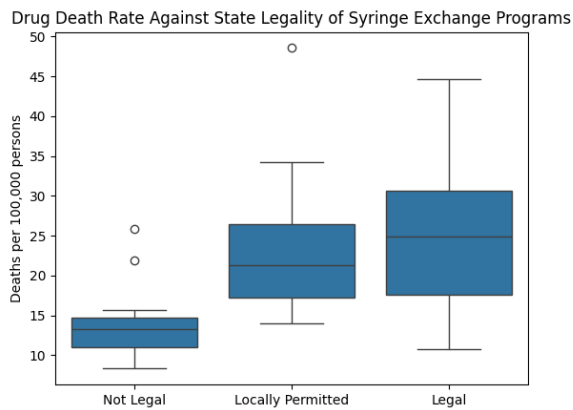


Figure 2: Drug-related death rate against state legality of syringe exchange programs, showing a higher incidence of deaths in states with SSP legality.

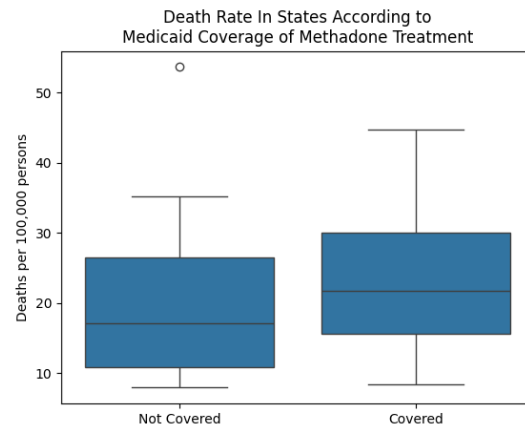


Figure 3: Drug-related death rate against state Medicaid coverage of methadone treatment, showing a higher incidence of deaths in states with covered methadone treatment.

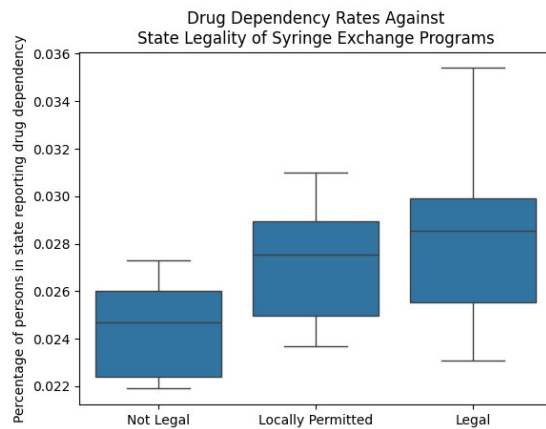


Figure 4: Drug-related death rate against state legality of syringe exchange programs, showing a higher incidence of deaths in states with SSP legality.

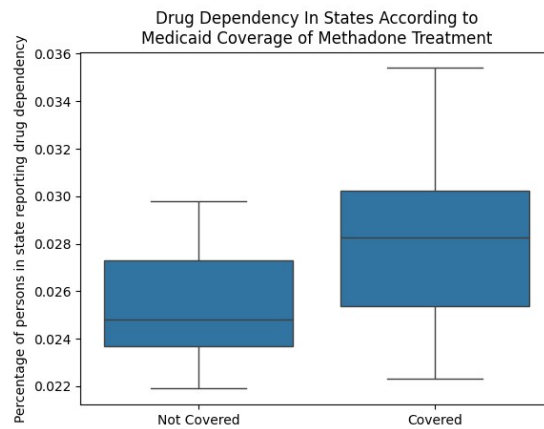


Figure 5: Drug dependency rate against state Medicaid coverage of methadone treatment, showing a higher incidence of deaths in states with covered methadone treatment.

While this makes sense in the context of the current literature, given that availability is likely to follow public health patterns (e.g. syringe exchange programs may be implemented

more robustly in states with known high rates of drug dependency), this does indicate that neither categorical variable will be a strong predictor of target outcomes. A more robust approach would be to track robust state-level data over time to compare improvements, rather than compare states against each other. However, this data is unavailable.

The most applicable predictor value for a given target variable is the *number* of syringe programs compared against drug-related deaths. However, the closest years for which this data is available is 2023 and 2019, respectively. (Comparisons between 2024 and 2019 show very similar results.) However, even this value lacks overwhelming predictive potential as, though the maximum death rate in states with higher numbers of SSPs is significantly lower, the *average* rate is only barely lower.

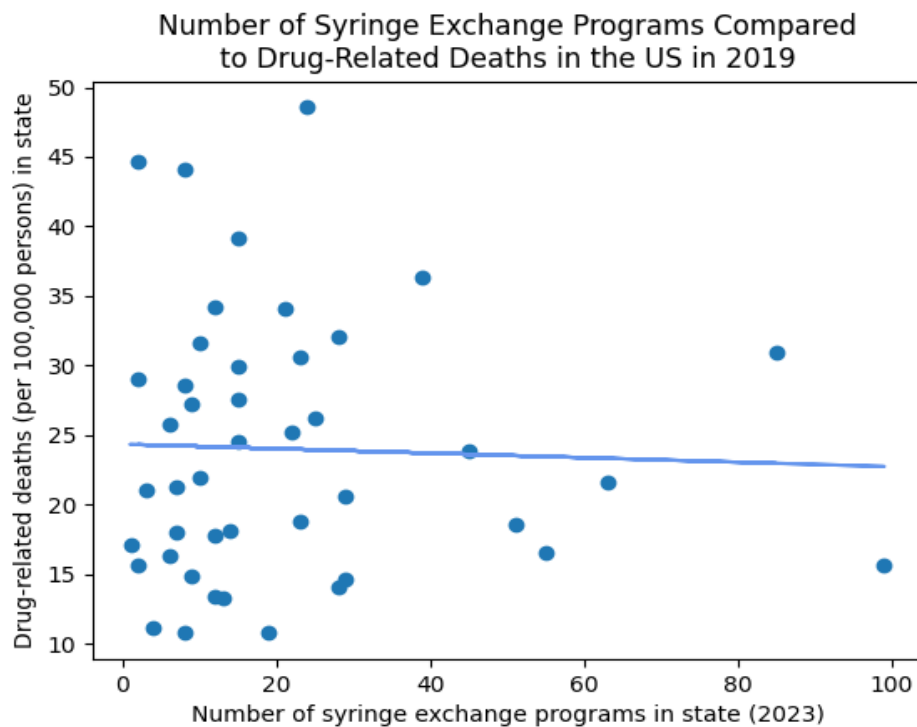


Figure 6: Number of syringe services programs (2023) compared to drug-related deaths (2019). While the maximum value over 50 programs is lower than the maximum value under 50 programs, the average is only marginally lower, with a slope of ~ 0.016 .

With this predictive variable as the strongest candidate for training a model, it would be irresponsible to train a model on any of these predictors, as it is likely that the training data is not sufficiently robust.

Conclusion

The available datasets are not sufficient to move forward with training a model. However, I believe that this preliminary research makes a strong case for specific improvements on data collection moving forward.

The largest weakness in the available data is the lack of comparable time-series data. For a given state, it would be extremely valuable to track outcomes year-over-year compared to probable predictor variables such as the number of available SSPs. Single-moment insights are inadequate to examine the impact of policy decisions and harm reduction resources.

Assumptions

Within the scope of this project, I have assumed that amFAR has archived and aggregated the most recently available data. I believe that this is a safe assumption given amFAR's up-to-date copyright and attribution and the fact that their most recent internal report was provided before the most recent available data, indicating long-term upkeep.

Limitations

The lack of locally-focused time-based data makes training a model based off of commonly-indicated predictors very difficult. Additionally, there is a hard limit on data in that the United States contains a maximum of 50 states (notwithstanding outlying territories), which is prohibitive for a comprehensive model. This could be addressed by using more dense data such as that which is collected on the county level, but this data is not always available or readily possible to match with other variables.

Challenges

There are two core challenges of leveraging the available data: Capturing true time-series change and determining cause-and-effect relationships.

While time series data is available year-over-year for drug-related deaths, other outcome data, such as drug dependency rates, and their predictor values, such as syringe service program availability, are less available. Additionally, drug-related death data in the relevant format is only available until 2019, while syringe program availability data (as an example) becomes publically available in 2023. These represent a serious mismatch that prevent this analysis from being able to pinpoint any policy-related cause-and-effect relationships.

Relatedly, without this long-term data to indicate changes in health outcomes over time, the cause-and-effect relationship becomes muddled. For example, drug dependency is positively correlated with states whose Medicaid programs cover methadone treatment. The cause-and-effect relationship, according to available literature, indicates that methadone coverage has likely expanded *as a result of* increasingly negative health outcomes in the relevant states, but it is impossible to determine the true nature of this relationship without data over time.

Future Uses, Additional Applications, and Recommendations

I believe that the best possible use of this research is to advocate for a small-scale data collection intensive: for example, outreach to established harm reduction groups, such as the aforementioned NCHRC, to do a data collection intensive in a single county such as Wake County, North Carolina. Within the scope of a single locale, over the course of a decade or more, it would be helpful to collect detailed information on different types of drug-related interventions, various ongoing health outcomes, and availability of resources.

Implementation Plan

The next logical step is to take this assessed gap in data to the relevant groups, i.e. the North Carolina Harm Reduction Coalition, and advocate for a program which closes the gap. I have reached out to the coalition to ask for their internal data and prepare a next-step assessment based on their internally-available private data and organizational needs.

Ethical Assessment

The primary reason that I determined it was inadvisable to produce a potential model for this data is an ethical concern, specifically that the model as represented would likely inadequately account for cause-effect relationships and indicate predicted values directly counter to the understanding of contemporary research on harm reduction efforts. An ineffective model would, in this case, be worse than no model at all.

Otherwise, the primary ethical concerns of this project entail data sourcing and preparation. I have endeavored to use the most recently-available data available and edit it minimally during preparation, only removing values which were not available. I also collected all data from reputable sources, namely the CDC and professional advocacy groups.

References

- Pridgen, B. E., Bontemps, A. P., Lloyd, A. R., Wagner, W. P., Kay, E. S., Eaton, E. F., & Cropsey, K. L. (2025). U.S. substance use harm reduction efforts: a review of the current state of policy, policy barriers, and recommendations. *Harm Reduction Journal*, 22(1).
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